

22/11/19

Indian Institute of Engineering Science and Technology, Shibpur
Dual Degree (B.Tech-M.Tech) 3rd Semester Final Examination, November 2019

Subject: Mathematics-III(MA-301)

(All Engineering Branches)

Use separate answer script for each half

Time : 3 hours

Full Marks : 70

First Half

Answer any seven questions

1. State axiomatic definition of probability. Using this definition show that for any two events A and B :

$$P(\overline{A}B) = P(B) - P(AB)$$

[5]

2. For n events $A_1, A_2, \dots, A_n$ connected to a random experiment E , show that

$$P(A_1 A_2 \cdots A_n) \geq \sum_{i=1}^n P(A_i) - (n-1).$$

[5]

3. In an examination 60%, 70% and 75% students have passed in the subjects Physics, Chemistry and Mathematics respectively. Find the minimum percentage of students who have passed in all the subjects.

[5]

4. Show that the function

$$f(x) = \begin{cases} |x|, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

is a possible density function and hence find the corresponding distribution function. [5]

5. Find the mean and variance of the Poisson distribution.

[5]

6. A fair coin is tossed 400 times. Using normal approximation to binomial distribution, find the probability of obtaining (i) exactly 200 heads (ii) between 190 to 210 heads, both inclusive. Given that the area under standard normal curve between $z = 0$ and $z = 0.05$ is 0.0199 and between $z = 0$ and $z = 1.05$ is 0.3531.

[5]

7. State and prove Tchebycheff's inequality for a continuous random variable.

[5]

8. The probability density function a two-dimensional random variable (X, Y) is given by

$$f(x, y) = \begin{cases} k(x + y), & x \geq 0, y \geq 0, x + y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find k and evaluate $P(X < \frac{1}{2}, Y > \frac{1}{4})$.

[5]

9. Fit a straight line to the following data:

x_i	y_i
1	2.4
2	3
3	3.6
4	4
6	5
8	6

[5]

10. Define correlation coefficient between two random variables and show that it lies between -1 and +1 both inclusive.

[5]

11. In a partially destroyed Laboratory record of an analysis of correlation data, the following results only are legible: $\text{Var}(X)=9$.

Regression equations: $8X - 10Y + 66 = 0$, $40X - 18Y = 214$.

Find (i) the mean values of X and Y , (ii) $\rho(X, Y)$, (iii) σ_Y .

[5]

12. (i) Define unbiased estimator for a population parameter.

(ii) Prove that the sample mean $\bar{X}$ is an unbiased estimator of the population mean. [2+3]

13. Find the maximum likelihood estimate of μ for a Poisson population with parameter μ .

[5]

Second Half

Answer any **THREE** questions from this half
two marks are reserved for general proficiency in this half.

14. a) State and prove convolution theorem.

b) Use convolution theorem to find $L^{-1}\left\{\frac{1}{(p+2)^2(p-2)}\right\}$. [6+5=11]

15. a) Find the Laplace transform of the following :

(i) $f(t) = e^{-2t} \cos 3t + 5e^{-2t} \sin 3t$

(ii) $f(t) = \begin{cases} 3 & 0 \leq t < 5 \\ 10 & 5 \leq t < 8 \\ 0 & t \geq 8 \end{cases}$

b) Using Laplace transform, solve the equation $\frac{d^2y}{dt^2} + t \frac{dy}{dt} - y = 0$

if $y(0) = 0$, $\frac{dy}{dt} = 1$ for $t = 0$. [3+3+5=11]

16. a) Using Laplace transform, prove that $\int_0^\infty \left(\frac{\sin t}{t}\right) dt = \frac{\pi}{2}$.

b) Find the basic feasible solutions of the system of equations

$$x_1 + x_2 + x_3 = 8$$

$$3x_1 + 2x_2 = 18,$$

$$x_1, x_2, x_3 \geq 0.$$

[6+5=11]

17. a) Solve the following LPP by Simplex method :

$$\begin{aligned} &\text{Maximize } Z = x_1 + x_2 + 3x_3 \\ &\text{subject to } 3x_1 + 2x_2 + x_3 \leq 3 \\ &\quad \quad \quad 2x_1 + x_2 + 2x_3 \leq 2 \\ &\quad \quad \quad x_1, x_2, x_3 \geq 0. \end{aligned}$$

b) Prove that a hyperplane is a convex set.

[8 + 3]

18. Solve the following LPP using Charnes Big M -method :

$$\begin{array}{ll}\text{Minimize } Z = 2x_1 + 4x_2 + x_3 \\ \text{subject to } & x_1 + 2x_2 - x_3 \leq 5 \\ & 2x_1 - x_2 + 2x_3 = 2 \\ & -x_1 + 2x_2 + 2x_3 \geq 1 \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Or

Use Two Phase method to solve the following LPP :

$$\begin{array}{ll}\text{Maximize } Z = 2x_1 + x_2 + x_3 \\ \text{subject to } & 4x_1 + 6x_2 + 3x_3 \leq 8 \\ & 3x_1 - 6x_2 - 4x_3 \leq 1 \\ & 2x_1 + 3x_2 - 5x_3 \geq 4 \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

[11]
